Theoretical Analysis of Blockchain-based Proportional Fair Scheduling for O-RAN

***Abstract*—Blockchain-based proportional fair scheduling (BC-PFS) provides a decentralized and trustworthy platform for cross-operator resource sharing and user scheduling in O-RAN. In the BC-PFS, the trustworthy collaboration across operators highly rely on the blockchain consensus process, which may induce delayed scheduling and consensus failure. Yet the impact of blockchain consensus on the performance of BC-PFS in O-RAN remain unexplored. This paper establishes an analytical framework for BC-PFS and investigates the impact of delayed scheduling and consensus failure on system performance. We first develop a mathematical model of BC-PFS and prove the weak convergence of the throughput update process considering the delayed scheduling caused by relatively long block time. Moreover, we derive the semi-closed-form expression of user average throughput, along with its approximation and lower bound. We extend the model to evaluate throughput performance with both delayed scheduling and consensus failure. Furthermore, we reveal the trade-off between these two factors and derive the optimal scheduling interval. We also investigate the joint influence of channel coherence time and blockchain propagation delay on the optimal scheduling interval. The pooling effect is quantified by comparing the network utility of BC-PFS in O-RAN with that of conventional non-cooperative scheduling schemes. Finally, we conduct simulations to validate our proposed model and theoretical findings.**

***Index Terms*—Blockchain, O-RAN, proportional fairness, trustworthy, scheduling.**

I. Introduction

Open radio access network (O-RAN) emerges as a transformative paradigm characterized by openness, programmability, intelligence, and disaggregated architecture with interoperable interfaces [1], [2]. Through network function virtualization and open interfaces, O-RAN shifts the traditionally closed RAN architecture toward a more open environment that allows broader participation. As wireless resource sharing among multiple participants continues to increase [3]–[5], efficient and trustworthy coordination across networks or operators becomes increasingly necessary in O-RAN [6], [7]. However, the trust foundation for cross-operator user scheduling is often limited in O-RAN due to its openness and multi-stakeholder nature [8], [9]. Without trusted mechanisms, operators may have concerns about potential deceptions and become reluctant to share network resources and services with others [10]. These challenges highlight the pressing need for trustworthy scheduling paradigms that enable reliable collaboration in ~~multi-operator O-RAN~~.

To fill this gap, blockchain can be utilized as a critical enabler for establishing trust among multiple operators through immutable records and verifiable smart contract execution [11], [12]. Recent studies [13]–[15] have explored blockchain-enhanced inter-network collaboration to facilitate cross-operator coordination. Among them, the blockchain-based proportional fair scheduling (BC-PFS) is a representative approach for trustworthy cross-domain user scheduling. The BC-PFS provides an explicit trust mechanism required to enable reliable collaboration across heterogeneous networks in O-RAN. Specifically, smart contracts ~~execute user selection and throughput update on-chain~~. User selection follows the proportional fair rule, where the user with the largest ratio of its instantaneous rate to historical throughput is selected, and the throughput is subsequently updated according to the corresponding update rule. The scheduling result and updated throughput are then ~~confirmed through blockchain consensus and~~ recorded on a shared and immutable ledger. BC-PFS enhances inter-network trust for cross-operator scheduling by enabling verifiable scheduling execution and allowing ~~multiple~~ operators to maintain a consistent and traceable view of the scheduling process in O-RAN [16], [17].

However, ~~establishing trust~~ in O-RAN is not cost-free, ~~regardless of the underlying trust mechanism~~. In blockchain-enhanced cross-operator resource coordination and user association, trustworthy coordination largely relies on the consensus process, through which participating operators reach agreement on shared network states and coordination decisions [18]–[20]. In particular, BC-PFS represents a specific realization of such blockchain-enabled coordination for cross-operator scheduling in O-RAN. To ensure that scheduling decisions are made based on trusted historical throughput, the scheduling results and updated throughput should be validated through consensus before being used in later scheduling decisions. ~~Note that~~, the trust relationships between network operators highly rely on the blockchain consensus process~~. It~~ has a nonnegligible impact on the BC-PFS for O-RAN~~, which has been ignored~~.

On the one hand, the configured scheduling interval, or equivalently the block time, i.e., the average time to create a new block that contains the latest scheduling result, may be longer than the channel coherence time. In this case, the scheduling decision, made at the beginning of the scheduling interval based on the current user rates and historical throughput, remains fixed throughout the scheduling interval and may become outdated as the channel conditions vary. As a result, the mismatch between the scheduling decision and time-varying channel state, referred to as delayed scheduling, degrades the BC-PFS performance. Therefore, blockchains with short block time are more suitable for BC-PFS in O-RAN. For instance, Kaspa achieves a block time of 100 ms [21], while MegaEth and Yona reduce the block time to the order of 10 ms [22], [23]. Projects such as Magicblock and Final DeFi can generate blocks every 1 ms [24], [25], which approaches real-time operations. Although several blockchain platforms claim millisecond-level block times, reliably achieving such

short block time in practice remains challenging. Therefore, the block time may still be longer than the channel coherence time [26], [27]. This mismatch becomes more pronounced in high-mobility scenarios, where rapid fluctuations in channel state amplify the negative impact of delayed scheduling on the BC-PFS performance.

On the other hand, a short block time mitigates delayed scheduling but may leave insufficient time for consensus, increasing the risk of consensus failure for BC-PFS in O-RAN [28]. Specifically, the actual block propagation delay is variable, and blockchain maintainers may fail to complete consensus within the expected block time [29]. Such a failure prevents scheduling round, corresponding to a scheduling interval, from being completed. This leads to no effective user throughput during that round and degrades overall network performance in BC-PFS for O-RAN. Both delayed scheduling and consensus failure can degrade scheduling performance, while a trade-off exists between their impacts. It calls for a proper block time for BC-PFS in O-RAN to balance these impacts.

~~However~~, the performance of BC-PFS in Open RAN remains insufficiently understood, and the impact of the blockchain consensus process on it can not be quantitatively characterized. Meanwhile, how to select an appropriate block time to balance the delayed scheduling and consensus failure for BC-PFS in O-RAN is unexplored. Furthermore, blockchain plays a dual role in BC-PFS. The positive impact of the consensus process may offset gains from blockchain-enhanced trustworthy resource pooling. Yet the conditions under which BC-PFS can achieve a positive pooling gain remain unclear.

In this paper, we establish an analytical model for BC-PFS by considering the impact of consensus process. We reveal the trade-off between the delayed scheduling and consensus failure and examine their combined impact on user average throughput. Through this, we provide theoretical guidance for selecting an appropriate scheduling interval for BC-PFS and derive the conditions under which a positive pooling gain can be achieved. We quantify the pooling effect of O-RAN via BC-PFS. Overall, we provide a theoretical basis for the efficient use of BC-PFS in O-RAN. The main contributions of this paper are summarized as follows:

- We establish a mathematical model that captures the scheduling process across multiple operators for analyzing BC-PFS in O-RAN~~. It also considers~~ the impact of blockchain consensus process.
- We characterize the impact of delayed scheduling on BC-PFS. We prove the weak convergence of the scheduling process using stochastic approximation, and also derive the semi-closed-form expression for the average user throughput in O-RAN, along with its approximation and bounds when the scheduling interval tends to infinity.
- We further analyze the impact of blockchain consensus failure and reveal the trade-off between it and the delayed scheduling. We derive the optimal scheduling interval that balances these factors and examine how it varies with coherence time and blockchain propagation delay.
- We examine the pooling effect introduced by BC-PFS in O-RAN. We derive necessary and sufficient conditions for achieving a positive pooling gain with delayed scheduling and consensus failure.
- We validate our model under different network configurations through simulations. The results demonstrate the performance advantages of BC-PFS in O-RAN over non-cooperative scheduling despite the impact of blockchain consensus.

The rest of the paper is organized as follows. Section II establishes the O-RAN model. Section III presents the BC-PFS mechanism. Section IV analyzes the impact of delayed scheduling on BC-PFS performance. Section V considers the impact of consensus failure. Section VI identifies the optimal scheduling interval and investigates the trade-off between the delayed scheduling and consensus failure. Section VII evaluates the pooling gain. Section VIII provides simulation results, and Section IX concludes the paper.

II. O-RAN Model

A. Multi-operator O-RAN

We consider a multi-operator O-RAN comprising K heterogeneous networks. Each operator network serves N users, so the total number of users is KN, indexed by i $(i = 1, ..., KN)$. The architecture of O-RAN is illustrated in Fig. 1. O-RAN disaggregates base station functions into three components, namely O-RAN central unit (O-CU), O-RAN distributed unit (O-DU), and O-RAN radio unit (O-RU), which can be developed by different vendors or operators. Specifically, the O-CU performs higher-layer RAN functions, while the O-DU handles lower-layer functions, including radio resource scheduling. The O-DU connects to the O-RU, which supports radio transmission and reception. Moreover, O-RAN provides the near-real-time RAN intelligent controller (Near-RT RIC) and non-real-time RIC (Non-RT RIC) to enable programmable coordination and longer-term policy management, respectively. To enable trustworthy coordination among multiple operators, the BC-PFS introduces a two-layer blockchain architecture, where edge chains support local cross-operator user scheduling and the global chain facilitates coordination across entire network. The detailed architecture of BC-PFS in O-RAN is given in Section III.

Multiple operators leverage open and programmable RAN infrastructures and share a common frequency band through synchronized time-division multiplexing. The instantaneous rate of user i at time t, denoted as $r_i(t)$, is determined by the channel state of its serving operator at that time. We assume that the instantaneous rates $r_i(t)$ and $r_j(t)$ for any two distinct users $i \neq j$ are mutually independent.

B. User Scheduling in O-RAN

In O-RAN, users access the network through the O-RU and report their scheduling-related information. Based on the channel conditions and transmission demands, the O-DU performs user scheduling and allocates the radio resources through control signaling. User scheduling follows a time-division framework, where time is divided into uniformly spaced slots. In each slot, the shared spectrum resource is allocated to

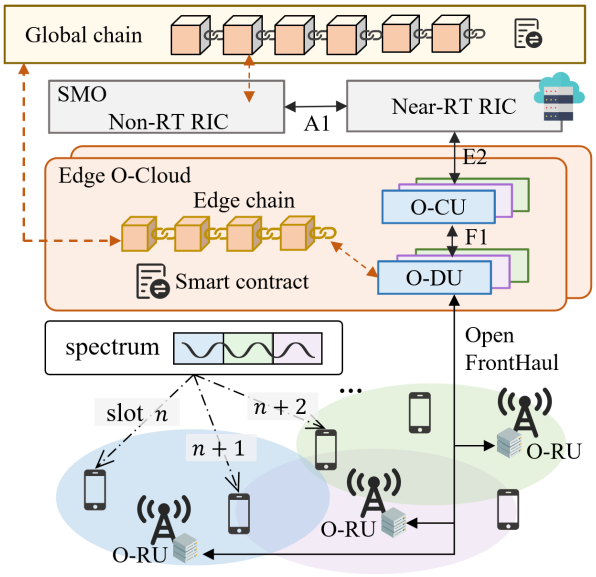


Figure 1. BC-PFS over multi-operator O-RAN.

only one operator, which serves one scheduled user over the bandwidth. This eliminates inter-operator interference and allows the scheduled user's transmission to occupy the entire bandwidth in that slot.

Each slot is defined as one scheduling interval of duration τ. The scheduling decision is made at time $t_n = n\tau$ and applied over the interval $[t_n, t_{n+1})$, which is referred to as slot n. User access to the frequency band is characterized by a binary indicator $I_i(t_n) \in \{0,1\}$, where $I_i(t_n) = 1$ indicates that user i receives transmission service during slot n. Consequently, the achievable rate for user i during slot n is given by $r_i(t) I_i(t_n)$. Furthermore, the throughput achieved by user i up to time t_n is denoted as $\mu_i(t_n)$. We define the long-term average throughput of user i, denoted by $\bar{\mu}_i$, as the limit of the expected instantaneous throughput over time, namely $\bar{\mu}_i = \lim_{n\to\infty} E[\mu_i(t_n)]$.

C. Wireless Channel Model

For user i, its instantaneous signal-to-noise ratio (SNR), denoted as $\mathrm{SNR}_i(t)$, can be modeled as an exponentially distributed random variable. The statistical characteristics of $\mathrm{SNR}_i(t)$ reflect the stochastic nature of channel quality fluctuations. Based on Shannon's theorem, the instantaneous transmission rate $r_i(t)$ of the user exhibits a logarithmic relationship with $\mathrm{SNR}_i(t)$. Let γ_i denote the mean SNR of the user i. Through statistical analysis, the mean $\bar{r}_i$ and variance σ_i^2 of the user transmission rate can be further derived as:

$$\bar{r}_i = \int_0^\infty \log_2(1+\gamma_i x)\exp(-x)\,dx, \tag{1}$$

$$\sigma_i^2 = \int_0^\infty (\log_2(1+\gamma_i x))^2 \exp(-x)\,dx - \bar{r}_i^2, \tag{2}$$

respectively. Here, $\bar{r}_i$ characterizes the long-term stability of the user transmission rate under time-varying channel conditions, while σ_i^2 captures its short-term fluctuations.

We define the channel coherence time T_c, during which the channel state remains stable. Beyond T_c, the channel parameters undergo significant variations. This provides a theoretical foundation for analyzing the coupling relationship between delayed scheduling and channel state, particularly in scenarios where blockchain scheduling intervals mismatch with channel coherence time.

III. Blockchain-based Proportional Fair Scheduling

In this section, we present the BC-PFS for O-RAN to enable efficient and trustworthy user scheduling across networks. BC-PFS executes the PFS rule through smart contracts and records scheduling states and allocation results on-chain, which enables verifiable scheduling execution and traceable resource allocation across operators. These capabilities enable the BC-PFS to support trustworthy collaborative user scheduling among multiple stakeholders in O-RAN.

As shown in Fig. 1, the BC-PFS in multi-operator O-RAN introduces a two-layer blockchain architecture. The edge chain, deployed at the edge O-RAN Cloud (O-Cloud), collects periodically reported user channel state information (CSI), executes scheduling algorithm via smart contract, and delivers scheduling results to the O-DU to coordinate resource allocation. It also immutably records scheduling states and allocation results, enabling the scheduling process to be traced and verified. Meanwhile, the global chain aggregates information from edge chains and supports overall coordination, cross-operator settlement, and policy optimization.

In principle, BC-PFS in O-RAN enables trustworthy collaborative scheduling across multiple operators. At each scheduling round, only one user is scheduled per time slot according to the proportional fairness rule. Mathematically, at time t_n, user selection follows the proportional fair rule:

$$i^*(t_n) = \underset{1\le i\le KN}{\operatorname{argmax}} \frac{r_i(t_n)}{\mu_i(t_n)}. \tag{3}$$

Accordingly, the user throughput is updated iteratively to reflect the latest resource allocations:

$$\mu_i(t_{n+1}) = (1-\alpha)\,\mu_i(t_n) + \alpha r_i(t_n) I_i(t_n), \tag{4}$$

where $\alpha \in (0,1)$ is the smoothing factor that determines the weights of historical throughput and newly achieved rate. Smaller α yields gradual throughput adjustments, while larger α enables faster responses to channel fluctuations. Under the proportional fair and throughput update rule, the long-term average throughput converges to the globally optimal solution of the network utility maximization problem [30]:

$$\text{maximize } U \triangleq \sum_{i=1}^{KN} \ln(\bar{\mu}_i). \tag{5}$$

The workflow of BC-PFS consists of the following four stages. 1) The workflow begins with registration, where users and operators are authenticated through the registration contract, with their verified identities recorded on-chain. 2) After registration, users periodically report their status, including key metrics like CSI and bandwidth requirements, to the O-DU.

Table I
SYMBOL DEFINITIONS.

Symbols	Explanations
K	Number of operator networks
N	Number of users per operator network
r_i	User rate
μ_i	User throughput
T_c	Channel coherence time
τ	Scheduling interval
T_p	Average blockchain propagation delay
U	Network utility

The O-DU then forwards the corresponding user states to the edge chain. The status reporting contract aggregates and records user rates for subsequent scheduling. 3) During each scheduling interval, the scheduling contract retrieves the latest user rates and historical throughput records. It then executes the PFS rule to determine the scheduled user and update the throughput records. The scheduling results are recorded in the new block and then delivered to the corresponding O-DU, which performs the radio resource allocation. 4) Finally, the edge chain aggregates scheduling service records and periodically synchronizes them with the global chain. The settlement contract calculates costs based on operator pricing rules and performs settlement among participants. The global information can be further used to support long-term policy optimization and parameter adjustment by the RIC.

Note that each scheduling round should be confirmed on-chain before the next round begins. This ensures that every scheduling decision is based on the trustworthy throughput state jointly validated by operators in O-RAN. However, the scheduling interval, determined by the block time, often exceeds the channel coherence time. The scheduling decision made at the beginning of the scheduling interval remains fixed during the interval despite channel variations, resulting in delayed scheduling and suboptimal decisions. Therefore, it is necessary to establish an analytical model that captures this mismatch and quantifies its impact on BC-PFS performance in O-RAN.

IV. DELAYED SCHEDULING MODEL

This section models the BC-PFS under delayed scheduling and conducts the weak convergence analysis of user throughput update process. We derive the semi-closed-form expression for user average throughput, together with its approximations and bounds under different limiting conditions. We have summarized important symbols in Table I.

A. Delayed Scheduling

To ensure that the scheduling decisions accurately capture the instantaneous channel state and achieve the optimal solution specified in (5), the scheduling interval τ is typically set to be smaller than T_c. However, in the BC-PFS, the scheduling interval τ may exceed T_c. This mismatch causes scheduling decisions to become outdated, which compromises the proportional fairness objective and causes degradation in network utility performance. When $\tau \geq T_c$, the channel state varies within a scheduling interval. This misalignment prevents the scheduling decisions from tracking channel dynamics in real-time, making throughput update based on suboptimal strategies that ultimately degrade long-term average performance. In this case, it is should depict this degradation under practical scheduling conditions.

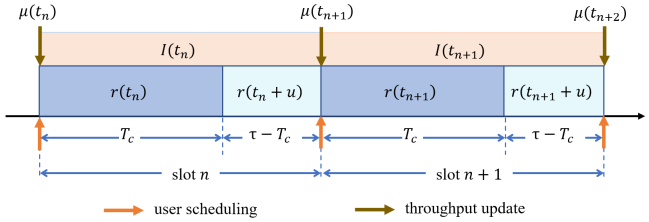


Figure 2. Illustration of BC-PFS delayed scheduling process.

The delayed scheduling process is illustrated in Fig. 2. During the n-th scheduling round, corresponding to time slot n, the scheduling decision $I(t_n)$ is made at t_n and remains fixed throughout the round. After t_n, the user rate r is assumed to remain constant for a duration of T_c and fluctuate thereafter due to channel variations. When the scheduling interval τ exceeds T_c, the channel may vary after $t_n + T_c$ while the scheduling decision $I(t_n)$ is unchanged, causing the decision to become outdated and no longer reflect the current channel state. This mismatch between the fixed scheduling decision and the varying channel state gives rise to delayed scheduling.

Since the scheduling decision is applied throughout the round, the service obtained by the scheduled user depends on the user rate over the entire scheduling interval. Therefore, the throughput update at t_{n+1} should account for the user rate variations within the round, and the throughput update rule for user i in (4) can be extended to the delayed scheduling case:

$$\mu_i^{\alpha}\left(t_{n+1}\right) = \tag{6}$$
$$(1-\alpha)\,\mu_i^{\alpha}\left(t_n\right) + \alpha\frac{1}{\tau}I_i\left(t_n\right)\left(T_c r_i\left(t_n\right) + \int_{T_c}^{\tau} r_i\left(t_n+u\right)du\right),$$

where μ_i^{α} is the sequence related to throughput update with smoothing factor α. The term $T_c r_i\left(t_n\right)$ represents the rate contribution during the period in which the channel remains unchanged, while $\int_{T_c}^{\tau} r_i\left(t_n+u\right)du$ captures the contribution under subsequent channel variations. The factor $\frac{1}{\tau}$ normalizes the accumulated user rate over the scheduling interval, yielding the average rate within the interval.

B. Weak Convergence Analysis

To characterize the long-term average throughput under delayed scheduling, we first analyze the weak convergence of the stochastic throughput update process in (6) using the stochastic approximation theory. The weak convergence analysis allows the asymptotic behavior of the stochastic process in (6) to be characterized by an associated deterministic ordinary differential equation (ODE), whose limit point captures the long-term average behavior of throughput. To establish this weak convergence, we introduce the following lemma based on the stochastic approximation results in [31]:

Lemma 1. *Under conditions (C1)-(C7), let $\{\theta_i\}$ be a discrete-time sequence indexed by $\varepsilon \to 0$, namely*

$$\theta_i^{\varepsilon}(t_{n+1}) = \theta_i^{\varepsilon}(t_n) + \varepsilon Y_i^{\varepsilon}(t_n). \tag{7}$$

For any nondecreasing integer sequence q_ε, there exists a subsequence such that the time-shifted process $\theta_i^{\varepsilon}(\varepsilon q_\varepsilon + \cdot)$ converges weakly to a continuous-time process $\theta_i(v)$ satisfying the integral equation:

$$\theta_i(v) = \theta_i(0) + \int_0^v \bar{g}_i(\boldsymbol{\theta}(s))\, ds, \tag{8}$$

where $\bar{g}$ denotes the averaged value of the dynamic process. The conditions are listed as follows:

(C1) $\{Y_i^{\varepsilon}(t_n); \varepsilon, t_n\}$ is uniformly integrable.

(C2) Define $\xi^{\varepsilon}(t_n)$ as the history information up to time t_n. There is a measurable function $g_i^{\varepsilon}(\cdot)$ and a random variable $\beta^{\varepsilon}(t_n)$ such that

$$E_t[Y_i^{\varepsilon}(t_n)] = g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n)) + \beta^{\varepsilon}(t_n), \tag{9}$$

where E_t denotes the expectation conditioned on the history information $\xi^{\varepsilon}(t_n)$.

(C3) $g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n))$ is continuous in $\boldsymbol{\theta}$.

(C4) $\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} E_t[\beta^{\varepsilon}(t_n)] = 0$ in the average sense.

(C5) For $\boldsymbol{\theta}$, the sequence $\{g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n)); \varepsilon, t_n\}$ is uniformly integrable.

(C6) For each $\delta > 0$ there is a compact set A_δ such that $\inf_{t,\varepsilon} P(\xi^{\varepsilon}(t_n) \in A_\delta) \geq 1 - \delta$.

(C7) For $\boldsymbol{\theta}$, there is a continuous-time function $\bar{g}_i(\cdot)$ such that the following expression holds in probability:

$$\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} E_t[g_i^{\varepsilon}(\boldsymbol{\theta}^{\varepsilon}(t_n), \xi^{\varepsilon}(t_n)) - \bar{g}_i(\boldsymbol{\theta})] \cdot I_{\{\xi^{\varepsilon}(t_n)\in A_\delta\}} = 0, \tag{10}$$

where $I_{\{\xi^{\varepsilon}(t_n)\in A_\delta\}}$ is the indicator function that equals 1 if $\xi^{\varepsilon}(t_n) \in A_\delta$ and 0 otherwise.

We next construct conditions (C1)-(C7) in Lemma (8) for the stochastic throughput update process in (6). Define the measurable function $Y_i^{\alpha}(t_n)$ as:

$$Y_i^{\alpha}(t_n) \triangleq \frac{1}{\tau} I_i(t_n)\left(T_c r_i(t_n) + \int_{T_c}^{\tau} r_i(t_n + u)\, du\right) - \mu_i^{\alpha}(t_n). \tag{11}$$

Then, we formulate the iterative sequence in (6) be expressed in the standard stochastic approximation form (7) as:

$$\mu_i^{\alpha}(t_{n+1}) = \mu_i^{\alpha}(t_n) + \alpha Y_i^{\alpha}(t_n), \tag{12}$$

where $\alpha \to 0$. First, we assume that $\{r_i(t_n), t_n < \infty\}$ is bounded. Therefore, there exists a constant M such that $\sup_{\alpha,t} E[|Y_i^{\alpha}(t_n)|] \leq M$. This implies that $\{Y_i^{\alpha}(t_n); \alpha, t_n\}$ is uniformly integrable, satisfying (C1).

Moreover, based on the wireless channel model established in Section II-C, $\{r_i(t_n), t_n < \infty\}$ constitutes a stationary stochastic process with time-invariant joint distributions. For each bounded random variable $r_i(t_n)$, let $\xi_i(t_n) = \{r_i(\ell) : \ell \leq t_n\}$ denote the historical rate information up to time slot n, and we define

$$h_i^{\alpha}(\boldsymbol{\mu}) \triangleq E_t\left[\frac{1}{\tau} I_i(t_n)\left(T_c r_i(t_n) + \int_{T_c}^{\tau} r_i(t_n + u)\, du\right)\right]. \tag{13}$$

Therefore, we have

$$E_t[Y_i^{\alpha}(t_n)] = h_i^{\alpha}(\boldsymbol{\mu}) - E_t[\mu_i^{\alpha}(t_n)]. \tag{14}$$

In this case, (C2) is satisfied with $\beta^{\alpha}(t_n) = 0$. Besides, since $\beta^{\alpha}(t_n) = 0$ for all conditions, (C4) is also met. Based on the Bayesian decomposition, (13) can be written as:

$$h_i^{\alpha}(\boldsymbol{\mu}) = \frac{1}{\tau}\Pr(I_i(t_n) = 1)\left(T_c E_t[r_i(t_n)|I_i(t_n) = 1] + E_t\left[\int_{T_c}^{\tau} r_i(t_n + u)\, du | I_i(t_n) = 1\right]\right). \tag{15}$$

With the scheduling rule in (3), $h_i^{\alpha}(\boldsymbol{\mu})$ can be expressed as

$$h_i^{\alpha}(\boldsymbol{\mu}) = \frac{1}{\tau}\Pr\left(\frac{r_i(t_n)}{\mu_i(t_n)} > \frac{r_j(t_n)}{\mu_j(t_n)}, \forall j \neq i\right) \cdot \left(T_c E_t\left[r_i(t_n) | \frac{r_i(t_n)}{\mu_i(t_n)} > \frac{r_j(t_n)}{\mu_j(t_n)}, \forall j \neq i\right] + E_t\left[\int_{T_c}^{\tau} r_i(t_n + u)\, du\right]\right). \tag{16}$$

The function $h_i^{\alpha}(\boldsymbol{\mu})$ satisfies local Lipschitz continuity in $\boldsymbol{\mu} \in \mathbb{R}_+^{KN}$. Similarly, owing to the boundedness and stationarity of $\{r_i(t_n), t_n < \infty\}$, $h_i^{\alpha}(\boldsymbol{\mu})$ is continuous in $\boldsymbol{\mu}$ and uniformly integrable. These properties guarantee (C3) and (C5). Meanwhile, according to [30], $\xi_i(t_n)$ can be assumed to reside in a compact sequence space, thereby satisfying (C6).

Leveraging the stationarity of $\{r_i(t_n), t < \infty\}$, we further define the steady-state expectation function for user i:

$$\bar{h}_i(\boldsymbol{\mu}) = \Pr\left(\frac{r_i}{\mu_i} > \frac{r_j}{\mu_j}, \forall j \neq i\right) \cdot \left(\frac{T_c}{\tau} E\left[r_i | \frac{r_i}{\mu_i} > \frac{r_j}{\mu_j}, \forall j \neq i\right] + \left(1 - \frac{T_c}{\tau}\right) E[r_i]\right), \tag{17}$$

where μ_i and r_i denote the canonical representations of $\mu_i(t)$ and $r_i(t)$ that exclude time dependencies, respectively. To establish the asymptotic equivalence between $h_i^{\alpha}(\boldsymbol{\mu})$ and $\bar{h}_i(\boldsymbol{\mu})$, we utilize the ergodicity of the stationary process. As t_n is sufficiently large, the dependence of $h_i^{\alpha}(\boldsymbol{\mu})$ on historical data approaches zero. By Khinchine's law of large numbers, for dependent stationary processes that satisfy the asymptotic independence, the average of conditional expectations converges to the steady-state expectation. Therefore, $h_i^{\alpha}(\boldsymbol{\mu})$ converges to the $\bar{h}_i(\boldsymbol{\mu})$ in probability:

$$\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} \left[h_i^{\alpha}(\boldsymbol{\mu}) - \bar{h}_i(\boldsymbol{\mu})\right] = 0. \tag{18}$$

Consequently, (C7) can be assured. In this way, all conditions of Lemma 1 on stochastic approximation are verified. The discrete-time throughput update process converges weakly to a continuous-time process of the form given in (8), from which the associated ODE is obtained by taking the time

derivative. The resulting ODE and its asymptotic behavior are characterized in the following theorem:

Theorem 1. *For BC-PFS under delayed scheduling, as $n \to \infty$, the user throughput $\mu_i(t_n)$ weakly converges to the limit point of the solution of the following ODE system:*

$$\dot{\mu}_i = \bar{h}_i(\boldsymbol{\mu}) - \mu_i, i = 1, \ldots, KN, \tag{19}$$

where $\bar{h}_i(\boldsymbol{\mu})$ is the conditional expectation of user rate r_i under event $\frac{r_i}{\mu_i} > \frac{r_j}{\mu_j}, \forall j \neq i$, with expression given in (17)*. Furthermore, employing ODE asymptotic analysis similar to [30], the limit point of* (19) *is unique and independent of initial conditions, which equals $\bar{\mu}_i$.*

Theorem 1 transforms the stochastic BC-PFS process into a deterministic differential equation for analysis. It demonstrates that $\mu_i(t_n)$ weakly converges to the unique asymptotically stable limit point of ODE $\bar{\mu}_i$ as $n \to \infty$. This provide a crucial mathematical foundation for the performance analysis of BC-PFS.

C. Throughput Analysis

This subsection derives the semi-closed-form expression for the long-term average throughput under delayed scheduling combing with the weak convergence property of (6). The resulting throughput characterization is given in the following theorem.

Theorem 2. *For BC-PFS, the approximation of the long-term average throughput of user i is*

$$\tilde{\mu}_i(\tau; T_c) \triangleq \frac{T_c}{KN\tau}\left(\bar{r}_i + \sigma_i g(KN, \eta_i)\right) + \frac{1-\frac{T_c}{\tau}}{KN}\left(\bar{r}_i \Phi(\eta_i) + \sigma_i \phi(\eta_i)\right). \tag{20}$$

where $\eta_i = \frac{\bar{r}_i}{\sigma_i}$ and $g(N, \eta_i) \triangleq N\int_{-\eta_i}^{\infty} x\phi(x)\Phi(x)^{N-1}dx$. The function $\phi(\cdot)$ and $\Phi(\cdot)$ are the probability density function (PDF) and cumulative distribution function (CDF) of the standard normal distribution, respectively. $\bar{r}_i$ and σ_i are the mean and standard deviation of r_i, given in (1) *and* (2)*.*

Proof. Please see Appendix A in the supplementary material. ∎

The first term in (20) represents the throughput contribution during the channel-stable period T_c, where the scheduling decision remains valid. In contrast, the second term corresponds to the remaining period $\tau - T_c$, where the channel varies while the scheduling decision remains unchanged.

When $\tau = T_c$, the channel state remains unchanged within the scheduling interval, and no delayed scheduling. Accordingly, the throughput result is consistent with that in [32] for the case without delayed scheduling. As $\frac{\tau}{T_c}$ increases from 1, the weight $\frac{T_c}{\tau}$ of the first term gradually weakens, while the weight $1 - \frac{T_c}{\tau}$ of the second term increases. This indicates that the fraction of scheduling interval during which the scheduling decision mismatch the varying channel state becomes larger, aggravating the impact of delayed scheduling on the average throughput.

We further simplify the user average throughput in (20) by consider the asymptotic case where the mean SNR of the user i satisfies $\gamma_i \to \infty$. This limiting condition allows us to derive semi-closed form expressions of throughput while preserving its main analytical characteristics.

Corollary 1. *When $\gamma_i \to \infty$, for user i in the BC-PFS, its average throughput can be approximated by:*

$$\tilde{\mu}_i^{\infty}(\tau; T_c) = \frac{1}{KN}\left(\bar{r}_i + \frac{T_c}{\tau}\sigma_i g(KN, \infty)\right). \tag{21}$$

Proof. Please see Appendix B in the supplementary material. ∎

The average throughput with delayed scheduling in the BC-PFS comprises two distinct components: the baseline throughput $\frac{\bar{r}_i}{KN}$ that reflects equal resource allocation among users, and multiuser diversity gain from channel variations. Notably, when $\tau = T_c$, the system achieves maximum scheduling gain equivalent to the collaborative PFS adopting accurate channel information. As τ increases, performance degradation due to delayed scheduling becomes more apparent and the multiuser diversity gain is progressively weakened.

Meanwhile, we consider a limited case $\tau \to \infty$, where nearly all scheduling decisions become suboptimal since they mismatch with the channel states. The corresponding average throughput is given in the following corollary.

Corollary 2. *When $\tau \to \infty$, for user i in the BC-PFS, its average throughput can be approximated by:*

$$\lim_{\tau \to \infty} \tilde{\mu}_i(\tau; T_c) = \frac{1}{KN}\left(\bar{r}_i \Phi(\eta_i) + \sigma_i \phi(\eta_i)\right). \tag{22}$$

Proof. Please see Appendix C in the supplementary material. ∎

When τ is sufficiently larger than T_c, the scheduling decision mismatches the current channel state during most of the scheduling interval, resulting in severe average throughput degradation. The result (22) is a limitation case of delayed scheduling and represents the lower bound of average throughput.

V. CONSENSUS FAILURE

Section IV has analyzed the performance degradation of BC-PFS caused by delayed scheduling, which can be mitigated by reducing scheduling interval τ toward T_c. However, when the scheduling interval τ becomes excessively small, the consensus process may fail to complete within a scheduling round, which results in consensus failure and undermines throughput efficiency. This section explores the degradation in the average throughput caused by consensus failures.

A. Probability of Consensus Failure

BC-PFS is implemented over the permissioned blockchain jointly maintained by participating operators. We introduce Raft as a representative consensus mechanism for the BC-PFS. In Raft, the consensus delay mainly consists of two message exchange phases, including the proposal propagation from the leader to followers and the acknowledgment collection from followers [33], [34]. When the actual consensus delay is longer than the predetermined scheduling interval τ, blockchain nodes may fail to receive critical messages in time.

This prevents the current scheduling round from completing successfully, which results in no effective service for users during that round and consequently degrades their average throughput. Since each message propagation delay can be modeled as an exponential random variable [35], we adopt a two-stage Erlang distribution with average propagation delay T_p to characterize consensus delay. In this case, the probability of timeout event, namely the probability of consensus failure, can be expressed as:

$$q_f = \exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right). \tag{23}$$

Consequently, the probability of successful scheduling is expressed as $1-q_f$. From (23), both the scheduling interval τ and the propagation delay T_p jointly influence the consensus failure probability q_f. Since $\frac{\partial q_f}{\partial \tau} = -\exp\left(-\frac{\tau}{T_p}\right)\frac{\tau}{T_p^2} < 0$, increasing the scheduling interval τ leads to a reduction in the consensus failure probability. A longer scheduling interval provides more time for nodes to achieve consensus and decreases the risk of failure. Additionally, since $\frac{\partial q_f}{\partial T_p} = \exp\left(-\frac{\tau}{T_p}\right)\frac{\tau^2}{T_p^3} > 0$, an increase in propagation delay T_p reduces the speed of information exchange among nodes and diminishes the efficiency of the consensus. This inefficiency increases the probability of consensus timeouts, which corresponds to the consensus failure probability.

B. Impact of Consensus Failure

To quantify the impact of consensus failure on user throughput, we extend the delayed scheduling model in (6) by incorporating the consensus success probability $1-q_f$ derived in Section V-A. Since a scheduling round contributes effective throughput only when consensus is successfully completed, the throughput update rule becomes:

$$\mu_i(t_{n+1}) = (1-\alpha)\,\mu_i(t_n) + \alpha(1-q_f)\frac{1}{\tau} \cdot I_i(t_n)\left(T_c r_i(t_n) + \int_{T_c}^{\tau} r_i(t_n+u)\,du\right). \tag{24}$$

This formulation explicitly couples the impact of blockchain consensus failure with delayed scheduling in the BC-PFS. Compared with (6), the throughput update rule in (24) only introduces the factor $1-q_f$, which is independent of user rate and throughput. Therefore, it retains the same stochastic approximation structure as (6), and the weak convergence property and the uniqueness of throughput limit point remain valid. Based on Theorem 2, we obtain the user average throughput under both delayed scheduling and consensus failure:

$$\tilde{\mu}_i^{\text{fail}}(\tau; T_c) = (1-q_f)\,\tilde{\mu}_i(\tau; T_c). \tag{25}$$

Meanwhile, based on Corollary 1, when $\gamma_i \to \infty$, (25) can simplify to

$$\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c) = \left(1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)\right) \cdot \frac{1}{KN}\left(\bar{r}_i + \frac{T_c}{\tau}\sigma_i g(KN,\infty)\right), \tag{26}$$

which demonstrates that average throughput is reduced due to consensus failure. The term $1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)$ reflects the impact of consensus failure, while $\frac{T_c}{\tau}$ captures the throughput loss caused by delayed scheduling. Thus, the scheduling interval τ establishes a trade-off between delayed scheduling and consensus failure. Reducing τ mitigates delayed scheduling but increases the risk of consensus failure.

VI. Scheduling Delay versus Consensus Failure

Building upon the analytical model of BC-PFS in Section V, this section investigates the optimal scheduling interval that balances the trade-off between delayed scheduling and consensus failure for maximizing user average throughput in the BC-PFS. Further, we reveal the combined influence of two significant factors, i.e., channel coherence time T_c and blockchain propagation delay T_p, on the optimal scheduling interval.

A. Scheduling Interval Optimization

In this subsection, we optimize the scheduling interval τ to maximize user average throughput in (26), which considers both delayed scheduling and consensus failure . Specifically, the optimization problem can be formulated as

$$\max_{\tau} \quad \tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c). \tag{27}$$

To solve this optimization problem, we first analyze the properties of the objective function $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ with respect to τ. The following theorem proves the existence and uniqueness of the optimal solution to maximizing $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$.

Theorem 3. *The function $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ has a unique stationary point $\tau^\star$ that satisfies $\partial\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)/\partial\tau\,|_{\tau=\tau^\star} = 0$ and achieves its global maximum at $\tau^\star$.*

Proof. Please see Appendix D in the supplementary material. ∎

Theorem 3 proves that the optimization problem in (27) admits a unique optimal solution $\tau^\star$, which is the stationary point of $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$. Moreover, the trend that $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ first increases and then decreases with τ reveals the trade-off between delayed scheduling and consensus failure. Specifically, increasing the scheduling interval τ has two opposite effects: it aggravates delayed scheduling by causing the scheduling decisions to rely more on outdated channel state information, while reducing the consensus failure probability. Over the range $(0, \tau^\star)$, as τ increases, the performance gain from mitigating consensus failure outweighs the loss caused by increased delayed scheduling, and thus the average throughput $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ increases with τ. At $\tau = \tau^\star$, these two effects are balanced, and $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ reaches its maximum. Conversely, in the interval $(\tau^\star, +\infty)$, further increasing τ yields only limited additional reduction in consensus failure probability as it gradually approaches zero, whereas the throughput loss caused by delayed scheduling becomes more significant. Therefore, $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ transition into monotonically decreasing with τ. Moreover, when $T_p \ll \tau$, the consensus failure probability is negligible, and thus $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ is mainly

affected by delayed scheduling and decreases monotonically with τ. The above analysis identifies the theoretical optimum $\tau^\star$. Since our delayed scheduling model assumes $\tau \geq T_c$, the feasibility of $\tau^\star$ should be further examined. If $\tau^\star \geq T_c$, then $\tilde{\mu}_i^{\text{fail},\infty}(\tau; T_c)$ attains its maximum at $\tau = \tau^\star$. Otherwise, the maximum is attained at the boundary $\tau = T_c$.

B. Impact of Blockchain Consensus Process

To better understand how T_c and T_p influence the optimal scheduling interval $\tau^\star$, we derive the following theorem.

Proposition 1. *The optimal scheduling interval $\tau^\star$ is strictly decreasing in the channel coherence time T_c and strictly increasing in the blockchain propagation delay T_p.*

Proof. Please see Appendix E in the supplementary material. ∎

Proposition 1 characterizes the monotonic relationship between the optimal scheduling interval $\tau^\star$ and the parameters T_c and T_p. First, as T_c decreases, the channel varies faster, the scheduling decision is outdated for a larger portion of the scheduling interval τ. In this case, reducing τ for more timely scheduling provides relatively limited throughout improvement, whereas a larger τ can effectively mitigate consensus failure. The impact of consensus failure becomes more significant in determining the optimal scheduling interval $\tau^\star$, leading to an increase in $\tau^\star$ as T_c decreases. Second, as T_p decreases, the consensus process can be completed within a shorter interval, which weakens the necessity of increasing τ to avoid consensus failure. The impact of consensus failure on the throughput optimization is less significant, while the delayed scheduling becomes the main factor. Therefore, the optimal scheduling interval $\tau^\star$ shifts toward a smaller value to mitigate the throughput degradation caused by delayed scheduling.

These results reveal that changes in T_c and T_p modify the trade-off between delayed scheduling and consensus failure, thereby shifting the optimal scheduling interval $\tau^\star$. Faster channel variation, i.e., smaller T_c, makes shortening τ to reduce sheduling delay less effective in improving the average throughput compared with enlarging τ to mitigating consensus failure. Meanwhile, a larger blockchain propagation delay T_p increases the necessity of a larger τ. These analysis provides insights into the variation of $\tau^\star$ under different network conditions and helps understand the trade-off between delayed scheduling and consensus failure in practical BC-PFS for O-RAN.

VII. Pooling Effect with delay

In this section, we systematically evaluate the pooling effect introduced by BC-PFS in O-RAN through network utility analysis. Furthermore, we derive the conditions for achieving positive pooling gain by balancing the benefits of blockchain-enhanced collaboration against the impact of the consensus process.

For the BC-PFS, multi-operator are jointly scheduled over the shared resources, serving a total of KN users. We define the network utility of BC-PFS in multi-operator O-RAN as:

$$\tilde{U}_{\text{coop}} = \sum_{i=1}^{KN} \ln\left(\tilde{\mu}_i^{\text{fail}}(\tau; T_c)\right). \tag{28}$$

In the non-cooperative PFS scenario where operators schedule independently, there are K isolated networks, each serving N users. The operators schedule independently with real-time channel tracking, i.e., $\tau = T_c$, and the scheduling process is unaffected by consensus failure. Based on (20), the average user throughput in these isolated networks is given by $\frac{1}{N}(\bar{r}_i + \sigma_i g(N, \eta_i))$. However, the effective spectrum availability per user is reduced by a factor of K due to time-division frequency sharing across networks, thus the throughput for each user scales as $\frac{1}{K}$. Therefore, the network utility of non-cooperative PFS is given by

$$\tilde{U}_{\text{iso}} = \sum_{i=1}^{KN} \ln\left(\frac{1}{KN}(\bar{r}_i + \sigma_i g(N, \eta_i))\right). \tag{29}$$

Combining with the network utility analyzed in (28) and (29), we define the pooling gain as the network utility gap between BC-PFS and non-cooperative PFS, given by

$$\Delta = \tilde{U}_{\text{coop}} - \tilde{U}_{\text{iso}} = \sum_{i=1}^{KN} \ln\left(\frac{KN\tilde{\mu}_i^{\text{fail}}(\tau; T_c)}{\bar{r}_i + \sigma_i g(N, \eta_i)}\right). \tag{30}$$

While the BC-PFS in O-RAN can enhance network aggregate and achieve pooling gain, delayed scheduling and consensus failure can diminish these gains. The following theorem gives the conditions under which the pooling gain can be positive.

Theorem 4. *The pooling gain is positive only if*

$$\frac{\frac{T_c}{\tau} KN g(KN,0) + \left(1 - \frac{T_c}{\tau}\right)\frac{1}{\sqrt{2\pi}}}{g(N,\infty)} > 1 - \exp\left(-\frac{\tau}{T_p}\right)\left(1 + \frac{\tau}{T_p}\right). \tag{31}$$

Moreover, a sufficient condition for positive pooling gain is

$$\frac{\eta_{\max} + \frac{T_c}{\tau} g(KN, \eta_{\max})}{\eta_{\max} + g(N, \eta_{\min})} > 1 - \exp\left(-\frac{\tau}{T_p}\right)\left(1 + \frac{\tau}{T_p}\right), \tag{32}$$

where $\eta_{\max} = \max_i \eta_i$ and $\eta_{\min} = \min_i \eta_i$.

Proof. Please see Appendix F in the supplementary material. ∎

Theorem 4 establishes the feasibility conditions for positive pooling gain introduced by BC-PFS in O-RAN. The necessary condition characterizes a requirement determined by the network scale and scheduling parameters, indicating that the benefit of cross-operator resource pooling must compensate for the performance loss caused by delayed scheduling. The sufficient condition provides a conservative guarantee for a positive pooling gain by considering the extreme user statistics. Specifically, the non-cooperative PFS is evaluated using the favorable extreme $g(N, \eta_{\min})$, whereas BC-PFS is evaluated using the unfavorable extreme $g(KN, \eta_{\max})$. This constructs a pessimistic comparison for BC-PFS. If the condition is satisfied even under this conservative combination, a positive

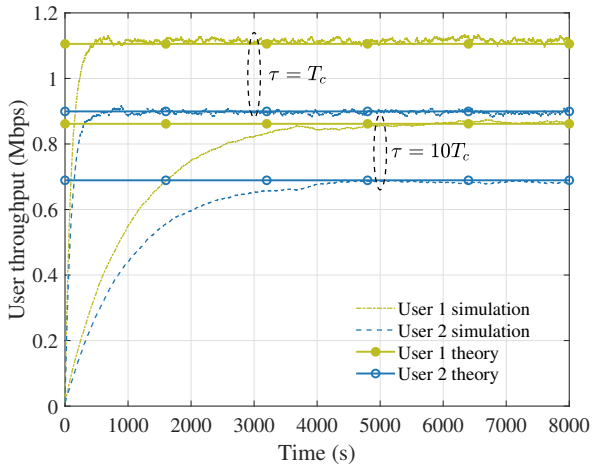


Figure 3. Simulation and theoretical results of user throughput with delayed scheduling.

pooling gain can be guaranteed. It also highlights the need to balance the pooling advantage against the losses caused by delayed scheduling and consensus failure.

VIII. Simulation

In this section, we present our simulation scenarios. We adopt stationary Rayleigh fading with constant and white external noise. The smoothing factor of throughput update is set to $\alpha = 10^{-4}$, which is small enough for the BC-PFS and acceptable measure of throughput. For all scenarios, we assume the bandwidth of 1 MHz.

To validate the analysis of BC-PFS with delayed scheduling presented in Section IV-C, we simulate a network with two users, each having average SNR values of 4.8 dB and 3 dB, respectively. We assume the channel coherence time $T_c = 10$ ms and examine two scheduling intervals, $\tau = T_c$ and $\tau = 10T_c$, to represent different levels of delayed scheduling. Fig. 3 presents the simulation outcomes and analytic results of Theorem 2 for both users. It can be observed that for a given α, the simulated throughput update consistently demonstrates stable convergence behavior across varying conditions. The convergence values of the simulation demonstrate a strong alignment with the asymptotic analytic results. Additionally, as $\frac{\tau}{T_c}$ increases, the negative impact of delayed scheduling information becomes more evident, which reduces user throughput.

Fig. 4 presents the throughput performance of a user under different SNR conditions in a network with a total of 2 users. It compares the high-SNR theoretical results from Corollary 1, alongside the standard theoretical values from Theorem 2 and corresponding simulation results. The percentage values annotated in the figures represent the relative errors between the high-SNR theoretical values and their corresponding standard theoretical values under each SNR condition. The results demonstrate that the high-SNR theoretical approximation has broad practical applicability. Although this approximation should strictly hold only under the condition when SNR tends to infinity, simulation results reveal that even at moderate SNR levels of 3 dB, the relative error remains below 1.2%. As the SNR increases, the relative error decreases further.

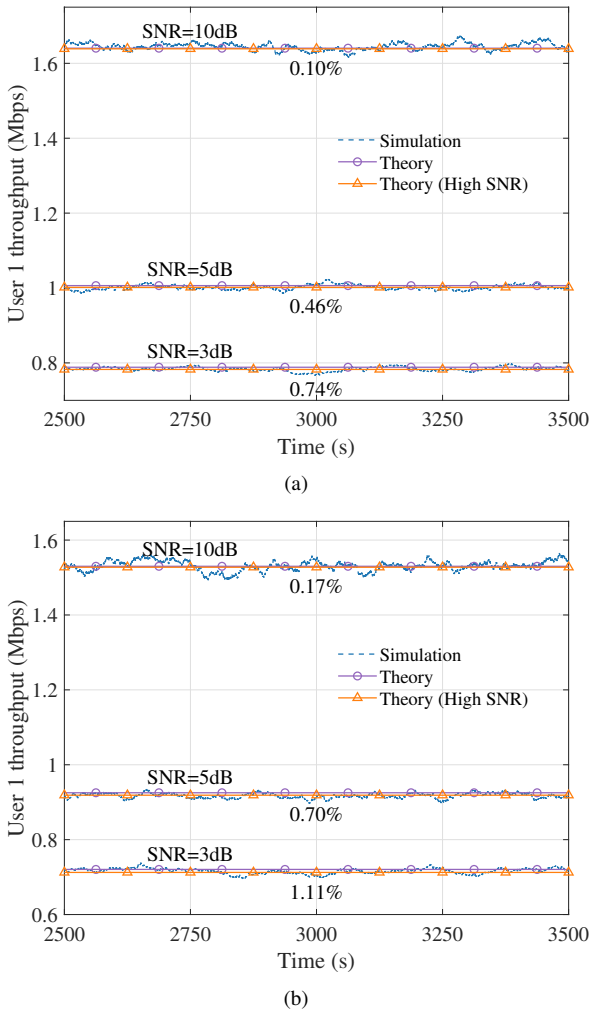


Figure 4. Simulation, theoretical results, and theoretical approximation of user throughput. (a) $\tau = 2T_c$. (b) $\tau = 5T_c$.

This phenomenon indicates that the high-SNR approximation possesses significant practical value in system design, and its applicability can reasonably be extended to moderate SNR regimes. Moreover, while increased delayed scheduling does amplify the discrepancy between theoretical values, its impact remains relatively limited. Comparing the cases of $\tau = 2T_c$ in Fig. 4(a) and $\tau = 5T_c$ in Fig. 4(b), it can be observed that under the same SNR conditions, the increase in delay contributes to an error growth of no more than 0.4%.

Furthermore, to validate the results of Corollary 2, Fig. 5 illustrates the relationship between the user average throughput and the ratio of scheduling interval to channel coherence time $\frac{\tau}{T_c}$. The average throughput corresponding to each ratio is calculated after the algorithm has converged. As shown in the figure, for smaller values of $\frac{\tau}{T_c}$, both users exhibit high initial throughput, which progressively decreases as $\frac{\tau}{T_c}$ increases. The throughput gradually converges toward the respective lower bounds as $\frac{\tau}{T_c}$ becomes larger. This trend highlights the increasing impact of delayed scheduling, where larger values of $\frac{\tau}{T_c}$ reduce user throughput performance. The figure confirms that the simulation results align closely with the theoretical results in Corollary 2, validating the analytic lower bounds as

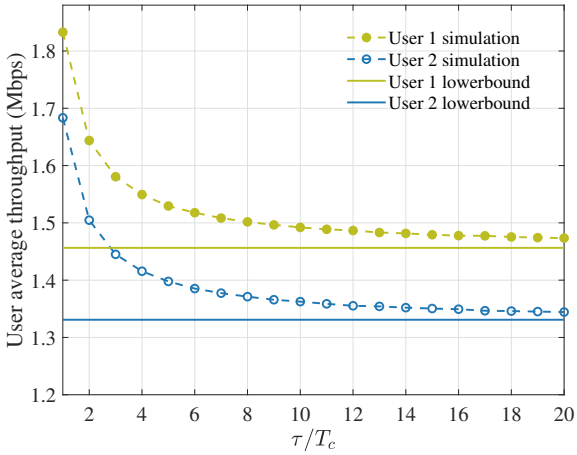


Figure 5. Simulation and theoretical lower bound of average throughput.

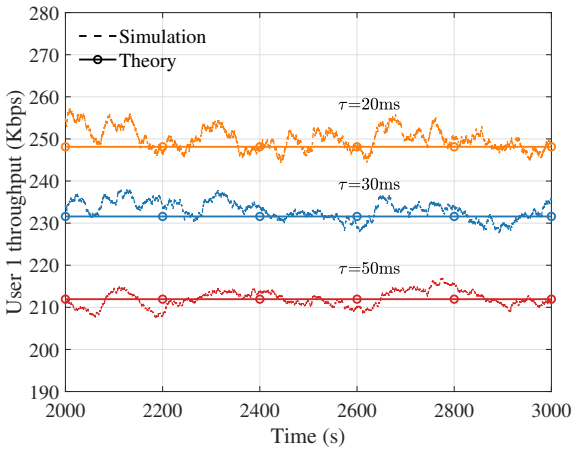


Figure 7. Simulation and theoretical results of user throughput with delayed scheduling and consensus failure.

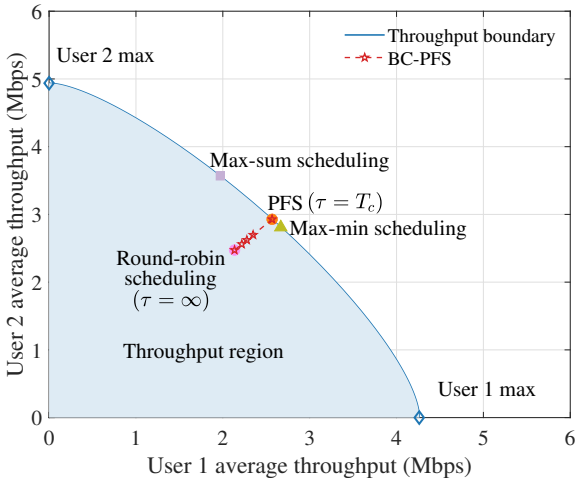


Figure 6. Throughput region: case of two users.

$\frac{\tau}{T_c} \to \infty$.

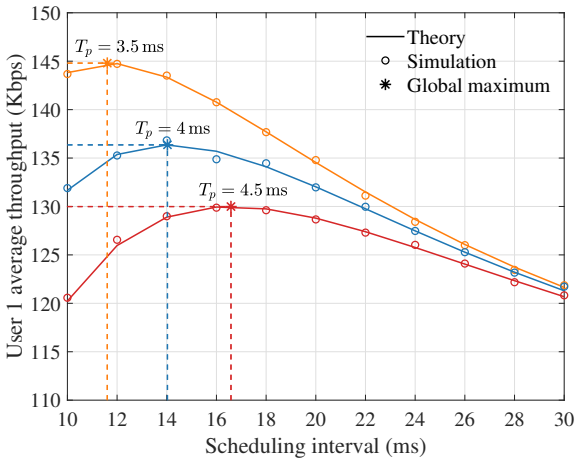
Fig. 6 presents the throughput region of a network with two users and the corresponding individual throughput achieved under different scheduling strategies. The dynamic scheduling is performed in each time slot, with only one user scheduled per slot. The figure highlights two extreme points that represent the average throughput each user would achieve if continuously scheduled over the entire duration, reflecting the maximum achievable throughput for each user. Given the Rayleigh fading channel settings, the maximum throughput corresponds to the mean user rate as defined in (1).

The maximum-sum scheduling, PFS, and max-min scheduling all operate on the throughput boundary, making these strategies Pareto optimal. That is, for any user, it is impossible to increase the throughput of one user without reducing the throughput of others [36]. Max-sum scheduling seeks to maximize the total network throughput by dynamically allocating resources based on each user's instantaneous rate, which tends to favor users with better channel conditions. In contrast, max-min scheduling prioritizes ensuring the user with the poorest channel conditions receives the most favorable scheduling, thereby maximizing the minimum throughput across all users. The PFS achieves an effective balance between fairness and overall system throughput. When $\tau = T_c$, the BC-PFS is equivalent to PFS. However, as τ increases, the individual throughput under the BC-PFS continuously decreases. When $\tau \to \infty$, the throughput collapses to the level of round-robin scheduling, which allocates resources equally among users regardless of their channel states.

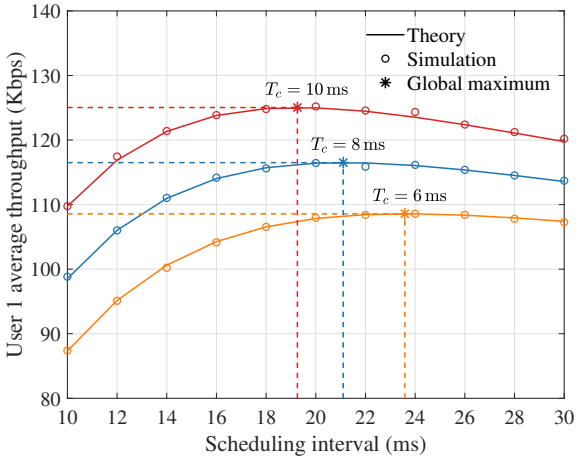
Furthermore, we present simulation results to validate the theoretical analysis of BC-PFS with both delayed scheduling and consensus failure in Section V. Fig. 7 illustrates the user throughput performance in a network of ten users, with the channel coherence time $T_c = 10$ ms and the blockchain propagation delay $T_p = 4$ ms. The simulated throughput closely converges to the theoretical results across all scheduling intervals, which confirms the robustness of the analytical framework.

This subsection evaluates the optimal scheduling interval concerning the impact of both delayed scheduling and consensus failure. Fig. 8 illustrates the average throughput performance of the network with $K = 2$, $N = 10$, where the numerically determined optimal scheduling interval for maximum throughput is explicitly identified. Fig. 8(a) examines the average throughput of one user under $T_c = 10$ ms for $T_p = 3.5$, 4, and 4.5 ms. The results show that the average throughput first increases and then decreases with longer scheduling intervals, reflecting the trade-off between wireless channel variations and blockchain propagation delay, which aligns with the theoretical analysis in Theorem 3. Moreover, as T_p increases, the optimal scheduling interval that maximizes throughput shifts toward larger values, consistent with Theorem 1. Concurrently, the average throughput declines due to the higher consensus failure probability induced by longer T_p, which exacerbates throughput loss.

Fig. 8(b) investigates the case where $T_p = 5$ ms under varying $T_c = 6$, 8, and 10 ms. A similar trend of initial increase followed by a decrease in throughput is observed. Furthermore, larger T_c reduces the optimal scheduling interval for peak throughput, again in agreement with Theorem 1. The average throughput improves simultaneously because a



(a)



(b)

Figure 8. Illustration of the optimal scheduling interval. (a) Under different T_p. (b) Under different T_c.

longer T_c increases the proportion of synchronized scheduling with accurate channel information. These findings provide valuable insights for parameter optimization in practical BC-PFS systems.

This subsection examines the pooling effect through the comparison of network utility under different scheduling methods and network settings. Fig. 9 presents the utility region of two operator networks, each serving two users with $T_c = 10$ ms and $T_p = 2$ ms. The results confirm that, when operators schedule independently within their respective networks, non-cooperative PFS yields superior utility for individual networks compared to max-sum and max-min fair strategies. This is due to the inherent proportional fairness properties of PFS that maximize logarithmic utility functions. Moreover, when the two networks are aggregated under the BC-PFS, the unified system achieves substantially higher overall utility, provided the delayed scheduling remains within appropriate bounds. This demonstrates that, by pooling resources and coordinating scheduling across multiple operators, the BC-PFS can enhance network performance beyond the capabilities of independent scheduling within each isolated network.

Moreover, Fig. 10 presents a comparison of the network

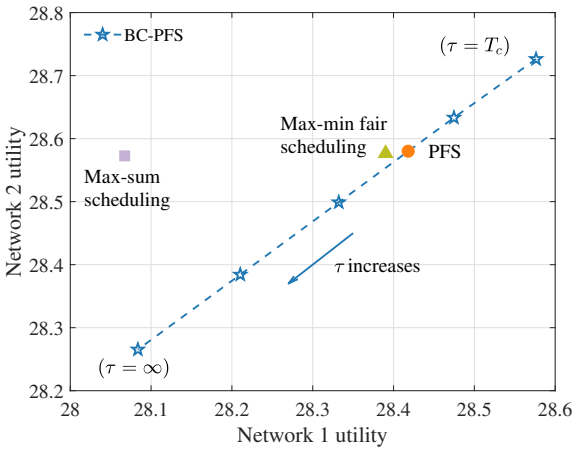
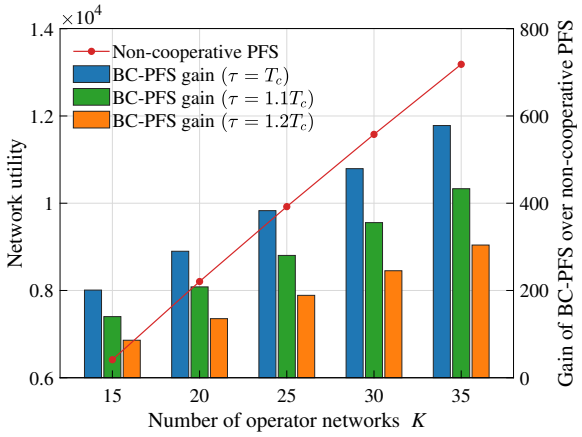


Figure 9. Utility region: case of two networks.

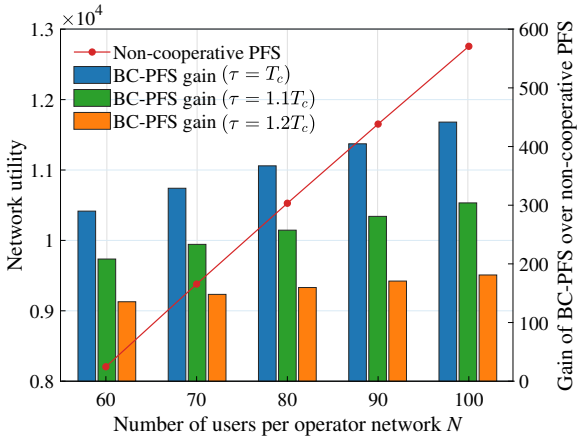
utility between the conventional non-cooperative PFS and the BC-PFS. The line represents the network utility of non-cooperative PFS, while the bars show the utility gain of BC-PFS over non-cooperative PFS under different scheduling intervals. Fig. 10(a) is evaluated under varying numbers of operator networks K with each maintaining sixty users. The BC-PFS corresponds to the network utility of the aggregated network, while non-cooperative PFS represents the sum of the utilities of multiple independent networks. The results indicate that for $\tau = T_c$, $\tau = 1.1T_c$, and $\tau = 1.2T_c$, the network utility under the BC-PFS is significantly higher than that under non-cooperative PFS. Moreover, as the number of operator networks grows, the advantage of the BC-PFS becomes increasingly pronounced. This demonstrates that the aggregation of multiple operator networks enhances the overall network utility, particularly as the network scale expands.

Fig. 10(b) maintains twenty operator networks with varying user density N. It is obvious that the BC-PFS consistently outperforms non-cooperative PFS across the case of acceptable delayed scheduling, which further validates the benefits of cooperative resource pooling. Consistent with the trend observed in Fig. 10(a), the network utility degrades as $\frac{\tau}{T_c}$ increases, reflecting the performance penalty caused by higher delayed scheduling. Meanwhile, when the user density grows, both scheduling methods exhibit improved network utility due to multi-user diversity gain, while the performance gap between them, namely the pooling gain ψ, also widens. This underscores the performance advantages of the BC-PFS in densely populated networks.

Furthermore, we evaluate the pooling effect of BC-PFS with delayed scheduling and consensus failure. Fig. 11 examines the relationship between network utility and scheduling intervals under the network configuration with $T_p = 4$ ms, $T_c = 10$ ms, and $N = 2$. The results demonstrate that the BC-PFS consistently outperforms non-cooperative PFS across different operator counts and scheduling intervals, validating the fundamental advantage of resource pooling among multiple operators. Notably, the network utility of the BC-PFS follows a similar non-monotonic pattern with respect to τ as the average throughput. As τ increases, the network utility first



(a)



(b)

Figure 10. Pooling gain of BC-PFS over non-cooperative PFS. (a) Under different K. (b) Under different N.

increases and then declines. The optimal performance occurs when the trade-off between wireless channel variations and blockchain propagation delays is balanced. These insights verify the theoretical analysis in Section VII and highlight the importance of carefully selecting scheduling intervals to enhance the performance of blockchain-enhanced O-RAN.

IX. Conclusion

In this paper, we analyze the impact of blockchain consensus process on the BC-PFS for O-RAN. We first develop an analytical model for BC-PFS considering delayed scheduling and prove the weak convergence of the user throughput update process. Based on the weak convergence property, we derive the semi-closed form expressions of the long-term average throughput and its approximation under high SNR or large scheduling interval. Consensus failures are further incorporated into the BC-PFS analysis framework. We uncover the trade-off between the delayed scheduling and consensus failure in the BC-PFS and examine the optimal scheduling interval that balances these two factors. Furthermore, we quantify the pooling effect and provide insights for the system parameter optimization. Simulation results support our analysis and illustrate the impact of various parameters on average throughput

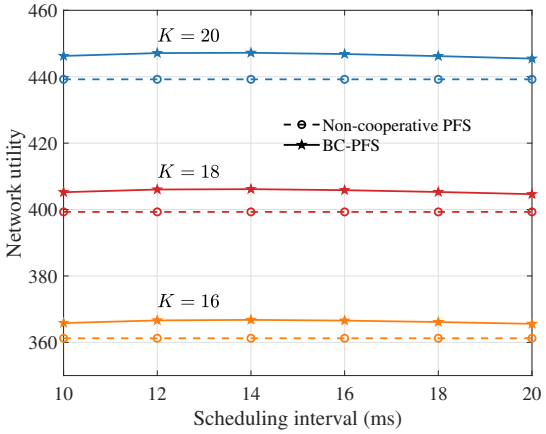


Figure 11. Comparison of network utility between non-cooperative PFS and BC-PFS under varying τ.

and network utility. This work provides crucial insights for system parameter configuration in practical implementations.

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